

Evaluation of fecal DNA preservation and extraction methods in Przewalski's horse

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Abstract The study compares the effects of three preservation methods (ethanol, drying and freezing) and two extraction methods (commercial kit and manual) on the PCR success rate and genotyping errors with fecal DNA from Przewalski's horse (*Equus przewalskii*). The interaction term between preservation and extraction was significant, and the highest amplification success rate (0.661) was obtained with samples stored in ethanol and extracted manually. The preservation method and time significantly impacted microsatellite amplification success rates. The method of preservation had an effect on the rate of allelic dropout (ADO), but the rate of false alleles (FA) varied nonsignificantly. Feces preserved in ethanol yielded the lowest ADO and FA, followed by drying and freezing. Our optimized preservation protocol probably have wide applicability to equids and rhinocerotids that produce large, high-fiber fecal pellets.

Keywords Noninvasive genetic sampling · Fecal DNA · Preservation · Extraction · Przewalski's horse

Noninvasive genetic sampling (NGS) has become indispensable tools for wildlife biologists. Technical improvements aim to reduce the genotyping error rates and thus

generate more reliable conclusions (Bonin et al. 2004; Pompanon et al. 2005; Waits and Paetkau 2005). However, there is mounting concern that genotyping errors may bias the results and negatively influence the management of threatened and endangered species (Taberlet et al. 1999; Fernando et al. 2003; Hoffman and Amos 2005). Efficient fecal DNA preservation has proved to be essential in recovering high quantities of high quality DNA, which increase the success rate of PCR and reduce genotyping error rates (Wehausen et al. 2004; Piggott 2004; Nsubuga et al. 2004; Hájková et al. 2006). However, no universally effective and applicable preservation method has been developed, and methods can be inconsistently successful across different species and studies (Wasser et al. 1997; Murphy et al. 2000, 2002; Frantzen et al. 1998; Piggott and Taylor 2003; Panasci et al. 2011).

Starting in 1985, Przewalski's horses were reintroduced to China. However, genetic monitoring is still insufficient for the captive and released populations because of the difficulty of obtaining adequate DNA samples. Here we evaluate three fecal DNA preservation methods and two extraction methods for their suitability for the analysis of reintroduced Przewalski's horses.

Twenty fresh fecal samples were collected. Each fecal sample was divided into three parts, and each part was preserved using one of the following three methods. (1) Drying, (2) Ethanol, (3) Freezing. Fecal DNA extraction was carried out using TIANamp Stool DNA Kit (TIANGEN, Beijing) and the manual method (see supplementary file 1). Four horse-specific microsatellite loci (HTG08, CA425, HMS07, VHL20; Shiue et al. 1999) were examined. Each 25 µL PCR mixture contained 1.5 mM MgCl₂, 2.5 µL 10× buffer, 0.2 mM dNTPs, 0.4 µM of each primer pair, 0.2 µL 10 mg/mL BSA, 0.4 µL *Taq* polymerase (5U/µL, TaKaRa) and 2 µL fecal DNA. PCR conditions were as follows: 5 min at 95 °C

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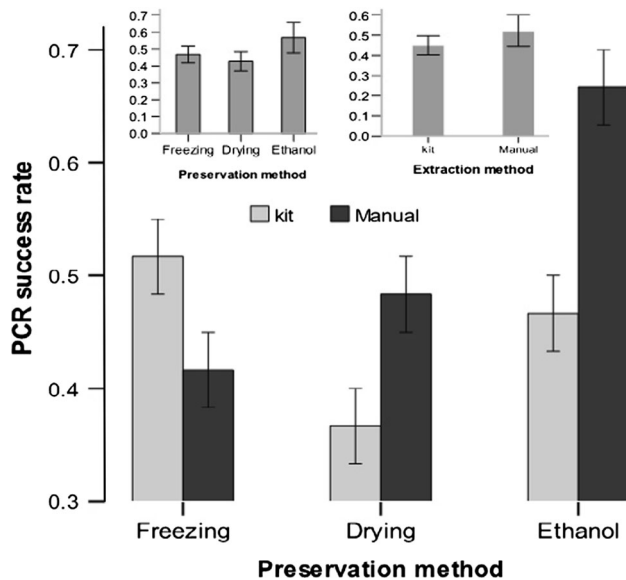
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Table 1 Experiment design of evaluation of fecal DNA preservation and extraction methods in Przewalski's horse

Fecal sampling	Preservation method	Preservation time	Extraction method	PCR success	ADO and FA
20 samples	Ethanol (E)	1, 2, 3, 6, 9 ^a	Manual (M) [1, 2, 3, 6, 9] Kit (K) [1]	EM [1, 2, 3, 6, 9] EK [1]	EM [1, 2, 3, 6]
20 samples	Drying (D)	1, 2, 3, 6, 9	Manual (M) [1, 2, 3, 6, 9] Kit (K) [1]	DM [1, 2, 3, 6, 9] DK [1]	DM [1, 2, 3, 6]
20 samples	Freezing (F)	1, 2, 3, 6, 9	Manual (M) [1, 2, 3, 6, 9] Kit (K) [1]	FM [1, 2, 3, 6, 9] FK [1]	FM [1, 2, 3, 6]

^a 1, 2, 3, 6, 9 indicate the feces preserved for 1 week, 2, 3, 6 and 9 months

**Fig. 1** Comparison of amplification success rates following different storage and extraction methods

followed by 35 cycles of 45 s at 95 °C, 1 min at the annealing temperature and 2 min at 72 °C, then 72 °C for 10 min. To optimize the combination of preservation and extraction methods, a two-way analysis of variance (ANOVA) was applied to ranked data (Dytham 2011).

We evaluated the effects of preservation time (1 week, 2, 3, 6 and 9 months) on PCR success rates using Cochran test and McNemar's test. Preservation effectiveness was evaluated after 1 week, 2, 3 and 6 months of preservation time by calculating allelic dropout (ADO) and false alleles (FA) using Wilcoxon's signed rank test (see experiment design in Table 1). Fecal genotypes were established by 8 % polyacrylamide gels based on the multiple-tubes approach (Taberlet et al. 1996). An allele was accepted only if it had been recorded at least twice (e.g. loci that gave rise to the same heterozygous genotype twice were accepted). In the ambiguous cases, a further four PCRs were performed to yield genotypes. Here, homozygote

consensus genotypes were scored only if the same allele was documented all seven times. Samples for which genotypes could not be assigned after seven positive PCRs were classified as 'unknown'.

For the six trials of combined preservation and extraction methods, the preservation method had a significant effect on the PCR success rate ($F_{(2,114)} = 7.234$, $P = 0.027$). Feces preserved with drying and freezing yielded lower PCR success rates than samples preserved with ethanol (Fig. 1). The interaction term between preservation and extraction was significant ($F_{(2,114)} = 22.111$, $P = 0.009$). In comparison, the highest amplification success rate (0.661) was obtained with samples stored in ethanol and extracted with the manual method, and the lowest for the drying/kit method (0.367).

The preservation method had a significant effect on the microsatellite PCR success rate ($\chi^2_4 = 44.31$, $P = 0.00$). Time-dependent decay in PCR success rate was observed for samples stored under all three preservation methods (ethanol, $\chi^2_4 = 11.652$, $P = 0.02$; drying, $\chi^2_4 = 12.696$, $P = 0.013$; freezing, $\chi^2_4 = 10.211$, $P = 0.037$). For ethanol preservation, reduction was observed at 9 months relative to 1 week ($P = 0.008$) and 2 months ($P = 0.025$). The PCR success rate for samples preserved by drying was reduced significantly at 6 ($P = 0.025$) and 9 months ($P = 0.014$). Freezing also displayed a time-dependent reduction in PCR success rate over time, with significant reductions observed at 6 ($P = 0.014$) and 9 months ($P = 0.020$), and between 2 and 9 months ($P = 0.034$). Regardless of preservation time, the type of preservation affected the rate of ADO ($\chi^2_5 = 12.338$, $P = 0.03$), but the rate of FA varied nonsignificantly ($\chi^2_5 = 10.725$, $P = 0.057$). Feces preserved with ethanol yielded the lowest ADO and FA, followed by drying and freezing for the four preservation times (Fig. 2). Feces preserved with ethanol yielded no significant increase of ADO from 1 week to 6 months preservation time ($P = 0.065$). Our results suggest that our optimized method may be suitable for large ungulate mammals such as equids and rhinocerotids produce large, high-fiber fecal pellets.

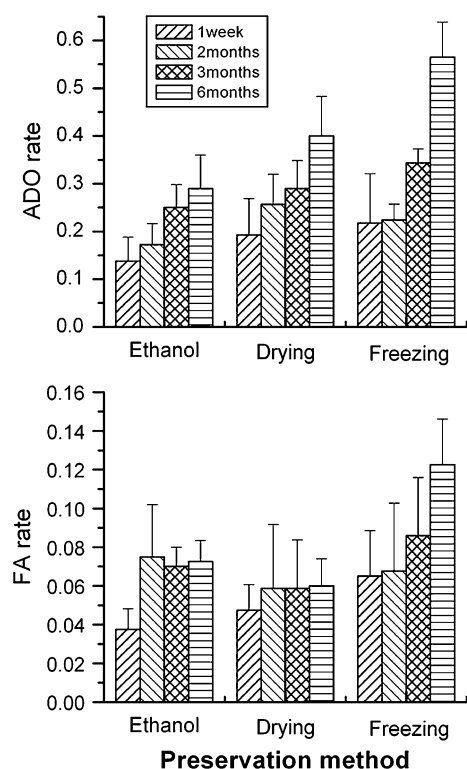


Fig. 2 Rates of ADO and FA of 4 microsatellite loci amplified from fecal DNA extracted from Przewalski's horse

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Conflict of interest The authors declare that they have no conflict of interest.

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